TIR = high denser – low denser : away from normal line

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G - 10 PHYSICS SECOND SEMESTER MID TWO **GINBOT 2018 E.C**

Topic: EM Waves And Geometrical Waves **Time Allowed: 1:00 hr**

Name: Bilanya Wondimagegnehu **Section:** 10th B **Number:** 12

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

false 1. The angle of incidence is always measured between the incident ray and the surface of the mirror.

True ② When light reflects off a surface, its frequency and wavelength remain unchanged. but in refraction frequency stays the same while wave varies

True 3. Light travels slower in water than it does in a vacuum.

low denser to high denser

false 4. When light enters a denser medium (like glass) from air, it bends away from the normal.

True ⑤ The refractive index of a material can never be less than 1.0.

false 6. Diffuse reflection does not follow the Law of Reflection because the surface is rough.

True 7. The speed of all electromagnetic waves is the same in a vacuum.

True ⑧ If light hits a boundary at an angle of 0° to the normal, it will not change direction, regardless of the medium.

II. Match the description in column A with part of EM wave spectrum list in column B.

A	B
E 9. used in night vision devices	A) Ultraviolet rays
C 10. used for satellite communication	B) Gamma rays
A 11. responsible for causing sunburn	C) Radio waves
B 12. primarily used to destroy cancer cells in medicine	D) Microwaves
D 13. has the longest wavelength and lowest frequency	E) Infrared waves
	F) Visible light

1 Radio – low frequency high wave len
micro
Infrared
visible
UV
α-ray
γ ray

III. Choose the correct one from the given alternatives and write the answer in the answer sheet in CAPITAL letter.

A 14. Which of the following describes an electromagnetic wave?

A) A transverse wave that can travel through a vacuum

B) A longitudinal wave requiring a medium

C) A wave consisting only of a magnetic field

D) A mechanical wave produced by sound

B 15. In an EM wave, the electric and magnetic fields are oriented:

A) Parallel to each other and the direction of travel

B) Perpendicular to each other and the direction of travel

C) At 45 degrees to each other

D) Only in the direction of the magnetic field

$\cos 30° = sin60° = \frac{\sqrt{3}}{2}, cos37° = sin53° = 0.8, cos45° = sin45° = \frac{\sqrt{2}}{2}, cos53° = sin37° = 0.6, cos60° = sin30° = 05$

VIBGYOR read-least bended

D 16. An FM radio station broadcasts EM radiation at a frequency of 100 MHz? wθ ?s A

A) 0.66 m B) 0.33 m C) 6.0 m D) 3.0 m

B 17. What is transferred by electromagnetic waves?

A) Mass B) Energy C) Medium particles D) Force only

A 18. In the visible spectrum, which color has the shortest wavelength?

A) Red B) Green C) Yellow D) Violet

B 19. What is the approximate wavelength range of visible light?

A) 1 nm to 100 nm C) 1 mm to 10 cm

B) 400 nm to 700 nm D) 700 nm to 1 mm

A 20. Which device operates using microwaves? r food

A) Radar system C) Radio station=Radio

B) Remote control-intraed D) Mirror

D 21. What is the frequency of a light with a wave length of 600 nm?

A) 200 MHz B) 500 MHz C) 2.0×10^{14}Hz D) 5.0×10^{14}Hz

C 22. When light passes from one medium to another, which property does NOT change?

A) Speed B) Wavelength C) Frequency D) Direction

D 23. Light travels fastest in:

A) Water B) Glass C) Diamond D) Vacuum

A 24. If a ray of light strikes a plane mirror at an angle of 25° with the mirror surface, what is the angle of reflection?

A) 65° B) 25° C) 50° D) 90°

B 25. Total Internal Reflection occurs only when light travels from:

A) A rarer medium to a denser medium C) Vacuum to air

B) A denser medium to a rarer medium D) One medium to another of the same density

C 26. The angle of incidence for which the angle of refraction is 90° is called the:

A) Normal angle C) Critical angle

B) Brewster's angle D) Angle of deviation

D 27. When a light ray strikes a mirror normally (perpendicularly), the angle of reflection is:

A) 90° B) 180° C) 45° D) 0°

C 28. Which of the following properties of light changes during reflection?

A) Speed B) Frequency C) Direction D) Wavelength

C 29. The refractive index of a medium is 2.0. What is the speed of light in that medium?

A) 3.0×10^8 m/s B) 6.0×10^8 m/s C) 1.5×10^8m/s D) 2.0×10^8m/s

D 30. The cause of refraction is the change in light's _______ as it enters a new medium.

A) Color B) Amplitude C) Phase D) Speed

D 31. Rainbows are formed due to a combination of:

A) Reflection and absorption

B) Interference only

C) Only simple reflection

D) Refraction, dispersion, and total internal reflection

A 32. What is the correct order of the EM spectrum in terms of increasing frequency?

low frequency high frequency

A) Radio, Microwave, Infrared, Visible, UV, X-ray, Gamma

B) Gamma, X-ray, UV, Visible, Infrared, Microwave, Radio

C) Radio, Infrared, Microwave, Visible, UV, Gamma, X-ray

D) Visible, UV, X-ray, Gamma, Infrared, Microwave, Radio

B 33. When light travels from water (n = 1.33) to air (n = 1.00), the ray will: denser to rare denser

A) Bend toward the normal C) Not bend at all

B) Bend away from the normal D) Stop moving

C 34. A pool of water appears shallower than it actually is because of:

A) Reflection B) Diffraction C) Refraction D) Dispersion

A 35. If the angle of incidence in air is 45° and the angle of refraction in a medium is 30°, what is the refractive index of the medium?

A) $\sqrt{2} = 1.414$ B) 1.5 C) 1.33 D) 0.707

B 36. "Diffuse reflection" occurs when light reflects off:

A) A smooth mirror surface C) A clear glass pane

B) A rough or irregular surface D) A still water surface

C 37. Which of these is an example of refraction?

A) Seeing your face in a lake C) A straw looking "broken" in a glass of water

B) Sunlight bouncing off a window D) A shadow forming on a wall

D 38. A dense glass has an index of refraction of $= 5/3 = 1.66$. What is the critical angle for total internal reflection for light leaving the glass and entering air for which$n = 1.0$?

A) 30° B) 60° C) 53° D) 37°

A 39. In reflection, the "angle of incidence" is the angle between the incident ray and the:

A) Normal line C) Reflected ray

B) Mirror surface D) Horizontal axis

B 40. In reflection, the incident ray, reflected ray, and the normal all lie:

A) In different planes C) Parallel to each other

B) In the same plane D) Perpendicular to each other

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Name: Biitanya Wondimagegnehu **Section:** 10th B **Number:** 12

1. false
2. True
3. True
4. false
5. True
6. false
7. True
8. True
9. E
10. CD
11. A
12. B
13. DC
14. A
15. B
16. D
17. B
18. ~~A~~D
19. B
20. A
21. D
22. C
23. D
24. A
25. B
26. C
27. D
28. C
29. C
30. D
31. D
32. A
33. B
34. C
35. A
36. B
37. C
38. D
39. A
40. B

37